



Amazon Route 53 — Complete Notes

Scalable DNS & domain registration · Connect names to resources

AWS Route 53 Service DNS Scope Global SLA 100%

A reader-friendly, all-in-one reference for Amazon Route 53.

Table of Contents

1. What is Route 53
2. Core Concepts
3. How DNS Resolution Works
4. Hosted Zones (Public vs Private)
5. DNS Record Types
6. Alias vs CNAME
7. Routing Policies
8. Health Checks & Failover
9. Domain Registration
10. Route 53 Resolver (Hybrid DNS)
11. TTL & Records Behaviour
12. Pricing
13. Common CLI Commands
14. Best Practices
15. Quick Mental Model
16. Common Interview Questions

1. What is Route 53

Amazon Route 53 is a **highly available, scalable DNS web service**. It does three core jobs: **register domain names**, **route internet traffic** to your resources by translating names (like `example.com`) into IP addresses, and **check the health** of your resources so traffic only goes to healthy endpoints.

 **Mental model:** Route 53 is the **phone book of the internet for your app** — it answers "what address does this name point to?" and can answer **differently** based on geography, latency, weights, or health. The name "53" comes from **DNS port 53**.

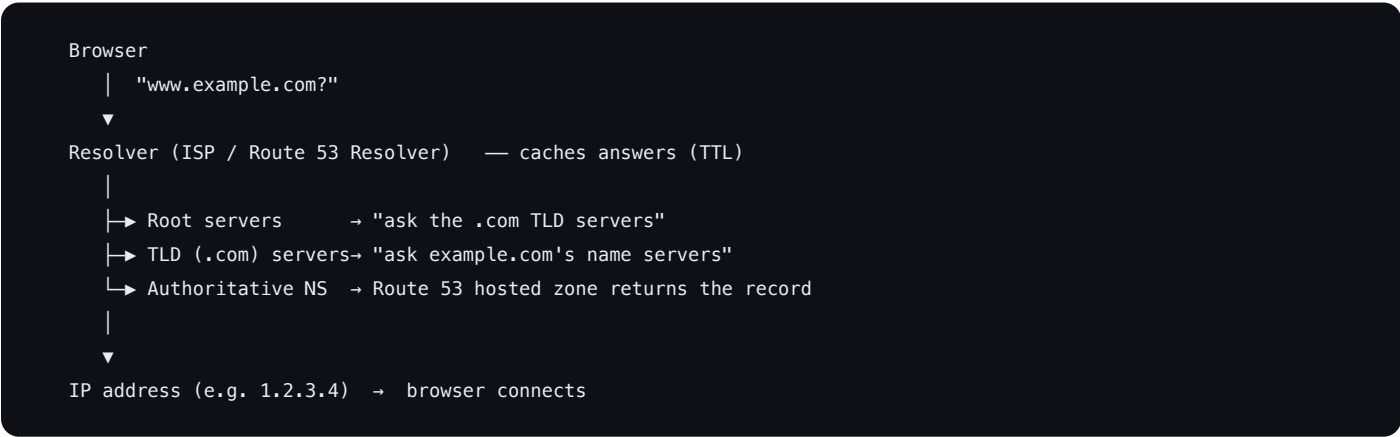
Key facts: **global** service (not region-scoped), backed by a **100% availability SLA**, and deeply integrated with AWS (ALB, CloudFront, S3, etc.).

2. Core Concepts

Concept	Description
Domain	A human-readable name you register (<code>example.com</code>).
Hosted Zone	A container for all DNS records of one domain.
Record (Record Set)	A single DNS entry (name → type → value), e.g. <code>www → A → 1.2.3.4</code> .
Name Server (NS)	The authoritative servers that answer for your zone.
TTL	Time-to-live — how long resolvers cache a record (seconds).
Routing Policy	The rule deciding <i>which</i> answer to return.
Health Check	A probe that tests whether an endpoint is healthy.
Resolver	The component that answers DNS queries (incl. hybrid on-prem).
Alias Record	A Route 53-specific record pointing to an AWS resource.

3. How DNS Resolution Works

When a user visits `www.example.com` , this chain resolves the name to an IP:



1. The **resolver** checks its **cache**; if expired/missing it walks the hierarchy.
2. **Root** → **TLD (`.com`)** → **authoritative name servers** (Route 53 for your zone).
3. Route 53 applies your **routing policy** (+ health checks) and returns the best answer.
4. The resolver **caches** the answer for the record's **TTL** and hands the IP to the browser.

4. 📁 Hosted Zones (Public vs Private)

A **hosted zone** holds the records for a domain. When you create one, Route 53 assigns **4 name servers (NS)**.

	Public Hosted Zone	Private Hosted Zone
Visible to	The public internet	Only inside associated VPC(s)
Use	Public websites, APIs	Internal service names, split-horizon DNS
Resolves from	Anywhere	Resources within the VPC

- For a **registered domain** to use Route 53, set the domain's NS records at the registrar to the **4 NS values** of its public hosted zone (automatic if you register the domain in Route 53).
- Split-horizon DNS**: same domain name resolves differently for internal (private zone) vs external (public zone) clients.

5. 🖨️ DNS Record Types

Record	Maps	Example
A	Name → IPv4 address	example.com → 1.2.3.4
AAAA	Name → IPv6 address	example.com → 2606:...
CNAME	Name → another name (alias)	www → example.com
Alias	Name → an AWS resource (Route 53 extension)	example.com → ALB/CloudFront/S3
MX	Mail servers (with priority)	example.com → 10 mail.example.com
TXT	Arbitrary text (SPF, DKIM, domain verification)	v=spf1 ...
NS	Name servers for the zone	delegation
SOA	Start of Authority (zone metadata)	one per zone
SRV	Service location (host + port)	SIP, etc.
PTR	IP → name (reverse DNS)	4.3.2.1.in-addr.arpa
CAA	Which CAs may issue certs	0 issue "amazon.com"

6. ⚖️ Alias vs CNAME

A frequent exam/real-world distinction:

	Alias (Route 53)	CNAME
Points to	AWS resources (ALB, CloudFront, S3, API GW, another Route 53 record)	Any DNS name
At the zone apex (<code>example.com</code>)?	✅ Yes	❌ No (DNS forbids CNAME at apex)
Cost	Free queries to AWS targets	Charged like a standard query
TTL	Managed by AWS (follows the target)	You set it
Resolves to	The target's current IP(s) automatically	Another name (extra lookup)

✅ **Rule of thumb:** pointing at an **AWS resource**? Use an **Alias** — especially at the **root/apex** domain where CNAME isn't allowed. Pointing at an external hostname? Use a **CNAME** on a subdomain.

7. 🧭 Routing Policies

Route 53 decides *which* record value to return based on the policy:

Policy	Returns the record based on...	Use case
Simple	One record, no logic	Single resource
Weighted	Split by assigned weights (e.g. 80/20)	A/B tests, gradual rollouts
Latency-based	The region with lowest latency to the user	Multi-region apps, fast UX
Failover	Primary if healthy, else secondary	Active-passive DR
Geolocation	The user's location (continent/country/state)	Localized content, compliance
Geoproximity	Distance to resources (with bias to expand/shrink)	Traffic-shaping by geography
Multivalued answer	Up to 8 healthy records at random	Simple client-side load spreading
IP-based	The client's IP/CIDR block	Route by known networks/ISPs

Most policies can be combined with **health checks** so unhealthy endpoints are dropped from answers.

8. 🩺 Health Checks & Failover

- **Health checks** probe an endpoint (HTTP/HTTPS/TCP) on an interval; a target is **unhealthy** after N failed checks.
- Types: **endpoint** checks (hit a URL/IP), **calculated** checks (combine other checks), and **CloudWatch alarm** checks (healthy = alarm OK).
- Tie a health check to a record so Route 53 **stops returning** unhealthy endpoints.

- **Failover routing** pairs a **primary** (returned while healthy) with a **secondary** (returned when primary fails) → active-passive disaster recovery.

```
Primary (health check OK)  ┌
                          │└─> Route 53 returns the healthy one
Secondary (standby)       └
```

9. 🛒 Domain Registration

- Route 53 can **register and manage domains** (acts as a registrar) for many TLDs — registration auto-creates a matching **public hosted zone**.
- Features: **auto-renew**, **transfer lock**, **privacy protection** (hides WHOIS contact info), and transfers in/out.
- You can also register elsewhere and just **host DNS** in Route 53 by pointing the registrar's NS records to Route 53's 4 name servers.

10. 🔗 Route 53 Resolver (Hybrid DNS)

- Inside a VPC, the **Route 53 Resolver** (the `.2` address) answers DNS automatically.
- For **hybrid** setups (cloud ↔ on-prem):
 - **Inbound endpoint** — lets on-prem servers resolve names in your private hosted zones.
 - **Outbound endpoint + Resolver rules** — forward queries for specific domains from the VPC to on-prem DNS.
- Enables one consistent DNS namespace across on-prem and AWS.

11. 🕒 TTL & Records Behaviour

- **TTL** controls how long resolvers cache an answer. **High TTL** = fewer queries (cheaper, but slow to change). **Low TTL** = fast propagation (good before a migration), more queries.
- **Lower the TTL ahead of a planned change** (e.g. to 60s) so cutover propagates quickly, then raise it back.
- **Alias** records to AWS targets don't take a manual TTL — they follow the target.
- DNS changes are **eventually consistent** across the internet's resolver caches (bounded by TTL).

12. 💰 Pricing

You pay for:

1. **Hosted zones** — per zone per month (first 25 cheaper; more is less each).
2. **Queries** — per million DNS queries (latency/geo/geoproximity cost a bit more than standard).
3. **Health checks** — per check per month (AWS-endpoint vs external, plus optional features like string matching/HTTPS).
4. **Domain registration** — annual fee per domain (varies by TLD).

 **Cost tips:** **Alias** queries to AWS targets are **free** (prefer Alias over CNAME for AWS resources) · consolidate records into fewer hosted zones · don't leave health checks running for retired endpoints · use sensible TTLs to limit query volume.

13. Common CLI Commands

```
aws route53 list-hosted-zones
aws route53 create-hosted-zone --name example.com --caller-reference $(date +%s)

# Create/update a record via a change batch file
aws route53 change-resource-record-sets --hosted-zone-id Z123 \
  --change-batch file://change.json

aws route53 list-resource-record-sets --hosted-zone-id Z123
aws route53 get-change --id /change/C123          # track propagation status

aws route53 create-health-check --caller-reference hc1 \
  --health-check-config Type=HTTPS,FullyQualifiedDomainName=example.com,Port=443

# Domains (registrar API)
aws route53domains register-domain --domain-name example.com --duration-in-years 1
aws route53domains list-domains
```

```
// change.json – upsert an A-alias to an ALB
{
  "Changes": [{
    "Action": "UPSERT",
    "ResourceRecordSet": {
      "Name": "example.com",
      "Type": "A",
      "AliasTarget": {
        "HostedZoneId": "Z35SXDO7RQ7X7K",
        "DNSName": "my-alb-123.us-east-1.elb.amazonaws.com",
        "EvaluateTargetHealth": true
      }
    }
  ]
}
```

14. Best Practices

- **Use Alias records** for AWS resources (free queries; works at the zone apex).
- **Lower TTLs before migrations**, restore them after, to control propagation speed.
- **Attach health checks** to routing records so traffic avoids unhealthy endpoints.
- **Latency or geolocation routing** for global apps; **failover** for DR.
- **Private hosted zones** for internal names; **split-horizon** when public + private differ.
- Enable **domain auto-renew + transfer lock + privacy** on registered domains.

- Keep DNS as **code** (Terraform/CloudFormation) and review changes — DNS mistakes are high-blast-radius.
 - Add **CAA** records to restrict which CAs can issue certificates for your domain.
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15. 🧠 Quick Mental Model

- **Route 53 = AWS DNS + domain registrar + health checks.** Global, 100% SLA, port 53.
 - A **hosted zone** holds your domain's **records**; **public** = internet, **private** = inside a VPC.
 - Resolution walks **root** → **TLD** → **your Route 53 name servers**, then caches by **TTL**.
 - **Alias** (AWS targets, free, works at apex) vs **CNAME** (any name, not at apex).
 - **Routing policies** pick the answer: simple, weighted, latency, failover, geolocation, geoproximity, multivalue, IP-based.
 - **Health checks** drop unhealthy endpoints; **failover** gives active-passive DR.
 - **Resolver endpoints** bridge DNS between on-prem and VPC (hybrid).
 - Lower **TTL** before changes; prefer **Alias** to cut cost; manage DNS **as code**.
-

16. ? Common Interview Questions

Rapid-fire questions interviewers ask about Route 53:

- **Q: Alias vs CNAME?** — Alias points to AWS resources, works at the **zone apex**, and queries to AWS targets are free. CNAME points to any name but **can't be used at the apex**.
 - **Q: What routing policies exist?** — Simple, Weighted, Latency-based, Failover, Geolocation, Geoproximity, Multivalue answer, IP-based.
 - **Q: How does Route 53 do failover?** — Health checks mark endpoints unhealthy; failover routing returns the primary while healthy, else the secondary (active-passive DR).
 - **Q: Public vs private hosted zone?** — Public resolves from the internet; private resolves only inside associated VPC(s). Same name in both = split-horizon DNS.
 - **Q: What does TTL control?** — How long resolvers cache an answer. Lower it before a migration for fast propagation, then raise it back.
 - **Q: How do you point a registered domain at Route 53?** — Set the domain's NS records (at the registrar) to the hosted zone's 4 name servers.
 - **Q: Latency vs Geolocation routing?** — Latency routes to the lowest-latency region; Geolocation routes by the user's physical location (compliance/localization).
 - **Q: Is Route 53 regional?** — No — it's a **global** service with a 100% availability SLA.
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 Notes compiled as a quick reference for Amazon Route 53.